

ShellSense: A Simple Keyboard Tool for Windows

Easy-to-read project abstract

Windows 10/11 • Python • PyQt6 • FastAPI • Firebase • Open source

Abstract

ShellSense is a fast and easy command tool for Windows. It helps people complete small computer tasks without opening many different apps. By pressing Ctrl + Shift + Space, users can open ShellSense from anywhere and type what they need in normal language. For example, they can calculate a number, convert units or money, find a file, copy an email address, set a timer, or run a safe system command. The main goal is to save time and reduce interruptions while working.

ShellSense uses a small machine-learning model on the computer to understand the user's request. This makes the main features fast and useful even when the internet is not available. It also includes a web dashboard for managing settings and snippets from anywhere. Important information, such as passwords and private notes, can be stored in an encrypted vault. The password stays on the user's device, and only encrypted data is saved online.

Problem

Windows users often stop their work to perform simple tasks. They may open a calculator, search the web for a conversion, check a different time zone, look through File Explorer, or retype the same information again and again. These small actions take time and break concentration. Passwords and other private details are also often kept in unsafe notes or browser fields. Existing launcher tools usually solve only one part of this problem, such as file search or app launching.

Solution

ShellSense brings these common tasks into one keyboard-first window. Users do not need to learn special commands. They can type requests such as "15% of 80," "100 dollars to rupees," "open last download," "copy my email," or "timer for 10 minutes." ShellSense reads the request, chooses the correct feature, and shows the result. Pressing Enter can copy the result and close the window, while Escape closes it without copying.

Main Features

Quick calculations and conversions: It handles maths, length, weight, temperature, speed, area, digital storage, and number systems.

Time and language help: It can convert time zones, check live currency rates, and translate text between several languages.

Snippets and files: Users can save common text, links, addresses, and file shortcuts. They can quickly open or copy recent downloads.

Timers and system tools: Users can set background timers, view startup programs, clean temporary files, and run selected Windows actions.

Secure vault: Private snippets are protected with PBKDF2 password protection and Fernet encryption. The master password is never sent to the cloud.

Web dashboard: Users can manage snippets, shortcuts, and commands through a simple web interface. Changes can be saved to Firebase and used by the desktop app.

How It Works

The desktop app is built with Python and PyQt6. A natural-language router sends each request to the right feature. A scikit-learn model helps identify the user's intent. FastAPI provides the web services, and Firebase Firestore stores synchronized settings. When the internet is unavailable, ShellSense can use local files as a backup. An optional AI assistant in the dashboard can help update selected settings through plain-language instructions.

Benefits and Future Work

ShellSense can reduce the time spent on everyday tasks. The project estimates that a user may save about six minutes each day by avoiding repeated app switching and typing. This could add up to about 36 hours in a year. These numbers are estimates and should be tested with real users in future studies. Future plans include support for macOS and Linux, browser and code-editor extensions, voice input, team snippet sharing, mobile access, and stronger enterprise security.

Conclusion

ShellSense is more than an app launcher. It is a simple personal command centre for Windows. It combines quick keyboard control, machine learning, useful daily tools, cloud settings, and secure storage in one place. Its open-source design makes it easy for students, developers, office workers, and other users to try and improve. The project shows how a clear interface and smart automation can make everyday computer work faster, safer, and easier.

Project: github.com/VRK1106/ShellSense • Live demo: shellsense.onrender.com